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The Palaeomagnetism of some of the Hawaiian Islands

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Summary

Palaeomagnetic directions of igneous rocks from five of the main Hawaiian islands are analysed. After partial demagnetization and the elimination of random sites, the directions are considered to be stable and isotropic and therefore are reliable indicators of the geomagnetic field in Hawaii during the last 5 million years. Both normal and reversed directions are found, which appear useful for stratigraphical correlation. The mean direction (357°, +30°) is close to that of the axial dipole field in Hawaii. This is consistent with palaeomagnetic data from other rocks of the same age and suggests that there has been no rotation of the Hawaiian islands since their formation. The scatter of site directions about the mean direction allows an estimate to be made of the maximum magnitude of secular variation. This suggests that secular variation has not been abnormal in Hawaii when considered over the last 5 million years, although more detailed sampling is required to determine the detailed behaviour of secular changes.

1. Introduction

The Hawaiian Archipelago consists of a chain of volcanic islands, reefs and shoals stretching some 2 600 km along a north-west–south-east arc. The main islands (Figure 1), which lie at the south-eastern end of the arc, are composed mainly of rapidly erupted "primitive" olivine-basalt lavas dipping at angles between 5° and 20° and often veneered with later, more acidic lava. The age of these islands was thought, on the grounds of morphology, raised beaches, dissection, etc., to decrease in general from north-west to south-east and to range from Recent to early or middle Tertiary (Stearns 1946). Recent radioactive dating and palaeomagnetic work (Tarling 1962; McDougall & Tarling 1963, 1964; McDougall 1963, 1964) have confirmed this general trend in age but have shown the rocks to be younger than previously thought and the samples vary in age from just over 5 million years (m.y.) to less than 1 m.y. (Table 1).

A collection of igneous rock samples was made from five islands, Maui, Lanai, Molokai, Oahu and Kauai, and the magnetic properties of these samples were studied in the laboratory. The survey is of a reconnaissance nature with the objective of investigating the gross behaviour of the Earth's magnetic field in the Hawaiian region during the last few million years. This is of interest as the present Earth's

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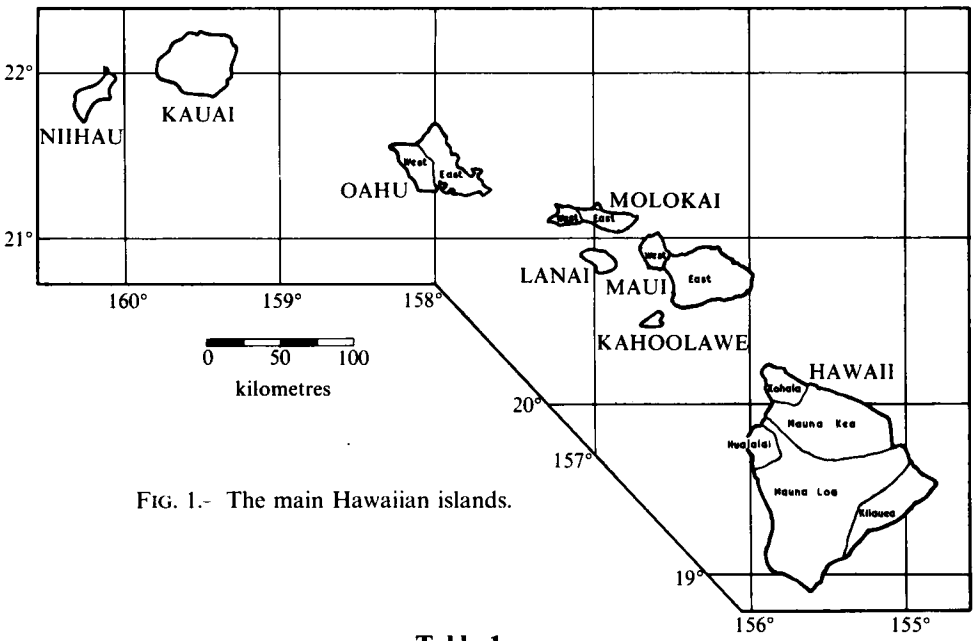
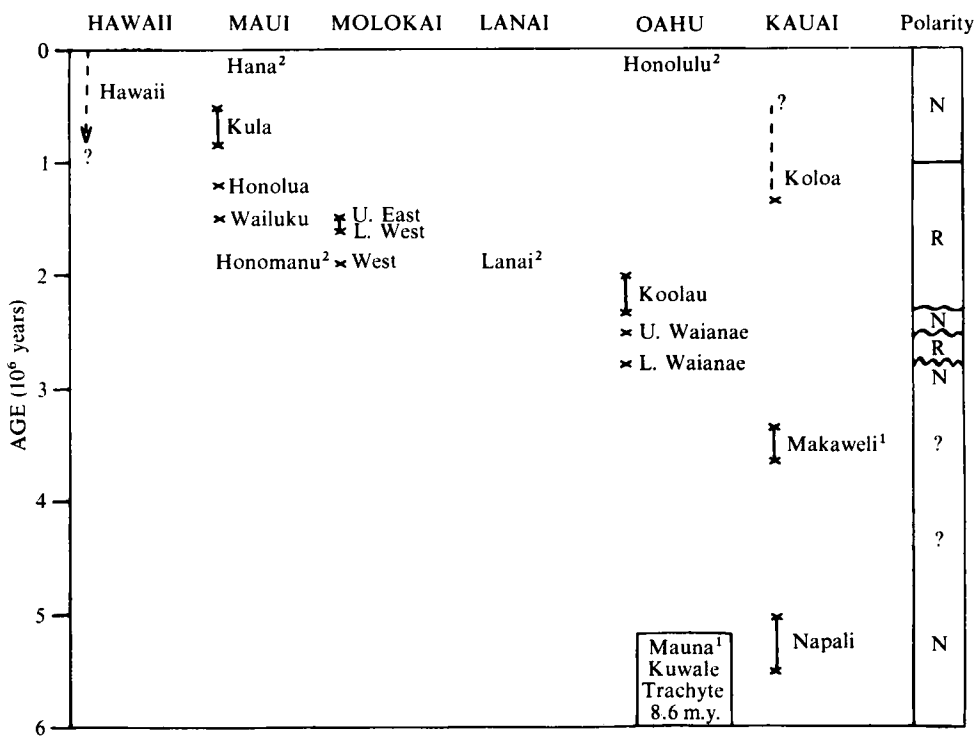


FIG. 1.- The main Hawaiian islands.

Table 1

Stratigraphical correlation of Hawaiian volcanic formations, based on geological, radio-active and palaeomagnetic data (McDougall & Tarling 1963)



N = normal. R = reversed. × × minimum range of age on radioactive evidence.

[1] Based on geological and radio-active data only.
[2] Based on geological and palaeomagnetic evidence only.

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magnetic field in the Pacific shows certain anomalous features when compared with the rest of the world. It is, on the whole, very close to the field of an inclined geocentric dipole (Cox 1962) and its secular change during the last 60 years has been very small (Fisk 1931). Palaeomagnetic observations on nine historical (1750 to 1955) lavas on the island of Hawaii (Doell & Cox 1963) are in agreement with magnetic records of early explorers in this region and suggest that the low rate of secular variation has persisted during at least the last 200 years, and earlier palaeomagnetic evidence (Doell & Cox 1961) suggests that it may have persisted throughout the Pleistocene. If secular variation has been permanently small in the Pacific area it implies the sources of secular change, probably eddy currents in the core (Lowes & Runcorn 1951; Alldredge & Hurwitz 1964), either consistently avoid this sector of the Earth, or that their effects are screened out, possibly by a region of high conductivity within the mantle beneath the Pacific (Runcorn 1956).

The average directions of natural remanent magnetization (NRM), after removal of low coercivity components in alternating magnetic fields, fall into two groups. One group (normal, N) is approximately parallel to the axial geocentric dipole field in Hawaii, and the other group (reversed, R) is antiparallel to it. The overall mean direction is close, but slightly displaced from the axial dipole field. The scatter of site directions about this average direction is thought to indicate a magnitude of secular variation during the last 5 m.y. which is consistent with models of the secular variation expected on a world-wide basis (Section 3.2). This suggests that although variations with periods of the order of 10^2 years may be absent or reduced in the Pacific region, variations on longer time scales are not anomalous in magnitude.

2. Methods

A total of 164 oriented hand samples of igneous rock was collected from 59 sites in 11 different rock series on five of the main Hawaiian islands. A site was a single lava flow or dyke from which samples were taken from a volume of rock which can be assumed to have acquired its thermo-remanent magnetization (TRM) over a short time. Two separate specimens, cylinders or disks, were drilled in the laboratory from most samples and the direction and intensity of their natural remanent magnetization (NRM) were measured by astatic magnetometers with an accuracy of 3° in direction and ± 5 per cent in intensity. Magnetic stability was studied with the alternating magnetic field method (Thellier & Rimbert 1954) using the apparatus and techniques described by Irving *et al.* (1961a).

Low coercivity components were removed from all specimens by treatment in 150 Oersted (peak) alternating field and the remaining intensity and direction of NRM measured. The directions of NRM, before and after treatment in alternating fields were analysed statistically to determine the average direction and to estimate the scatter of individual directions about this mean. Average directions were obtained by combining specimen directions graphically to get sample directions which were added vectorally, giving equal weight to each sample, to obtain the mean site direction. The length of the resultant site vector (**R**) was compared with 95 per cent significance point tables given by Watson (1956a) and Vincenz & Bruckshaw (1960). Sites in which **R** was not significant at this level, after partial demagnetization, were discarded as they were regarded as unstable or affected by lightning (Section 3.6), although if all the specimen directions had similar polarity the site was regarded as a reliable indicator of the polarity of the Earth's field, but not of

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its precise direction. The Fisher (1953) analysis was used, giving equal weight to each significant site direction, to obtain the overall mean directions and precision for various rock units.

The scatter of significant site directions, after partial demagnetization, about their mean was analysed in two ways. The precision of the overall mean direction was first estimated using the Fisher (1953) method (k), and secondly by the method given by Watson (1956b) and Watson & Irving (1957) in which allowance is made for the effect of within-site scatter (between sample directions at the same site) on the between-site scatter (β).

3. Results

3.1 *Geology and formation directions*

It is convenient to describe the geology and palaeomagnetic directions island by island before considering their overall characteristics of NRM. The formation directions are listed in Table 2 and the significant site directions, before and after partial demagnetization in 150 Oersted (peak) alternating magnetic field, are plotted in Figure 2. Only directions after partial demagnetization will be considered below. The range in age quoted for various formations is merely the limits of the age determined by McDougall (1963, 1964) for individual samples and do not necessarily give the full time range of that formation.

The Island of Hawaii (Stearns & Macdonald 1946) comprises the only active volcanoes in the Archipelago, Mauna Loa and Kilauea, and three inactive volcanoes,

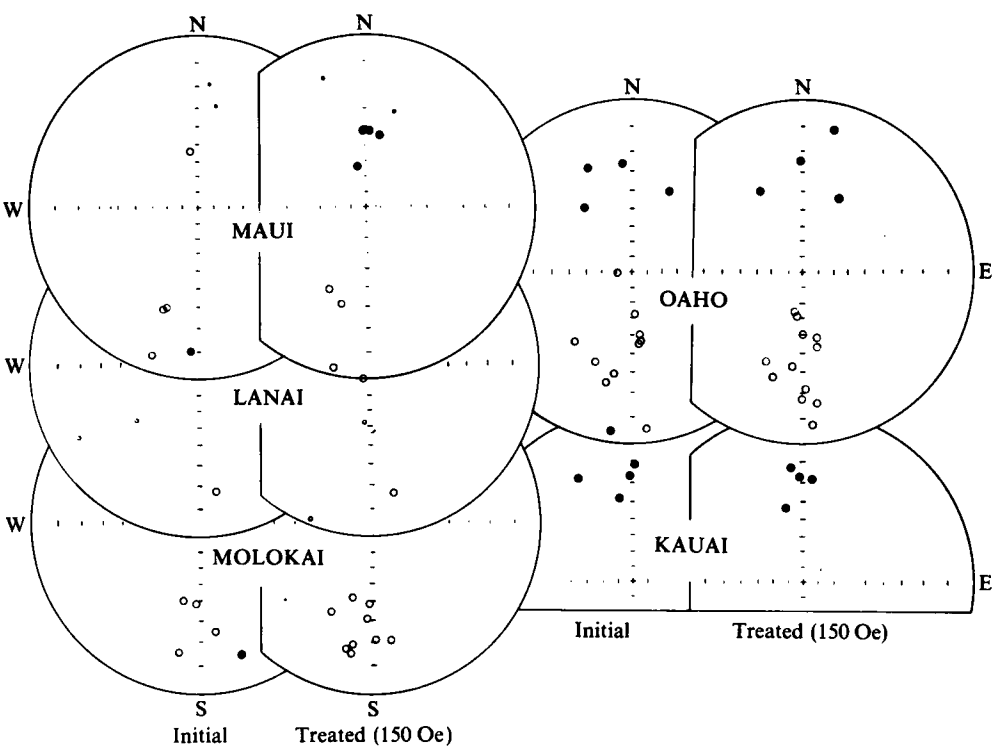


FIG. 2.—Stereographic projections of site directions, before and after treatment in 150 Oersted (peak) alternating magnetic field. ● Positive inclination; ○ negative inclination; ○ ● Significant sites: ∘∘ Sites with only one sample.

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Table 2

Formation and mean Hawaiian directions of NRM

Series or formation name	No. sites	No. samples	No. specimens	Initial D	Initial I	Initial **R**	Initial k	Treated (150 Oe) D	Treated (150 Oe) I	Treated (150 Oe) **R**	Treated (150 Oe) k	α	Pole Lat.	Pole Long.
Maui														
Hana Series	1	3	5	(9	+54)	—	—	(3	+42)	—	—	—	(86 N	125 W)
Kula Series	3	9	17	—	—	2·24	—	—	—	2·16	—	—	—	—
Wailuku Series	4	12	23	165	−12	3·84	18	168	−17	3·82	16	23	73 S	111 W
Honomanu Series	1	3	6	—	—	2·59	—	—	—	2·17	—	—	—	—
Lanai														
Lanai Series	1	2	3	(188	−17)	—	—	(190	−15)	—	—	—	(74 S	167 E)
Molokai														
East Molokai Series	6	16	29	164	−16	4·51	3	167	−24	5·27	7	28	75 S	101 W
Upper E. Formation	3	9	16	—	—	2·27	—	—	—	2·43	—	—	—	
Lower E. Formation	3	7	13	185	−31	2·97	74	179	−31	2·97	80	14	85 S	148 W
West Molokai Series	4	11	20	—	—	2·66	—	165	−25	3·91	35	16	74 S	94 W
Oahu														
Honolulu Series	2	6	11	(1	+27)	—	—	354	+17	1·97	35	43	76 N	49 E
Koolau Series	7	18	36	176	−31	5·99	6	179	−33	6·51	12	18	86 S	141 W
Koolau Basalt Formation	6	16	32	175	−38	5·20	6	179	−36	5·56	11	21	88 S	161 W
Koolau Dyke Complex	1	2	4	(173	+ 5)	—	—	(180	−17)	—	—	—	(77 S	157 W)
Waianae Series*	7	20	40	359	+38	6·59	15	358	+37	6·65	17	15	88 N	103 E
Waianae Basalt (R)	4	11	22	178	−38	3·79	15	177	−38	3·81	16	24	87 S	72 W
Waianae Basalt (N)	2	6	12	—	—	1·83	—	—	—	1·85	—	—	—	—
Waianae Dyke Complex	1	3	6	(170	−28)	—	—	(174	−32)	—	—	—	(83 S	103 W)
Kauai														
Koloa Series	1	3	6	(2	+27)	—	—	(3	+27)	—	—	—	(82 N	0)
Napali Formation	3	9	18	12	+28	2·92	25	4	+31	2·95	41	20	83 N	16 W
*All Hawaiian Sites**	38	106	303	—	—	—	—	357	+30	36·06	18	5	84 N	58 E

Declination (D) and inclination (I) in degrees from north and horizontal respectively. **R** is the length of the resultant vector, calculated giving equal weight to each site. k, α—estimate of precision (95 per cent) and semi-angle of cone of 95 per cent confidence (Fisher 1953). Pole position and mean direction only given when **R** is significant (95 per cent), and mean direction in parentheses when significance cannot be tested.

* Irrespective of polarity.

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Mauna Kea, Kohala and Hualalai, the latter being last active in 1800–1801. All rocks on this island are less than 800000 years old. Doell & Cox (1961) report that all samples they obtained from this island have normal magnetization.

Maui (Stearns & Macdonald 1942) is formed of two volcanoes, East and West Maui, connected by an isthmus formed mainly by lava flows from East Maui. East Maui is recognized as the younger and has been divided into three series. The oldest is the Honomanu Series, formed by basalts. The single site in this series is random, but is obviously reversed. The Kula Series (0·44 to 0·86 m.y.) conformably overlie the Honamanu Series and consist of thick andesitic olivine- and picrite-basalts. Two sites are random, and one site direction differs markedly from the three other normal sites in this formation. The divergent site is not obviously normal or reversed and, as there is some evidence of flow banding, this site direction is regarded as unreliable and is discarded. These rocks are overlain by the Hana Series which were erupted after a long quiescent period, and are variable in composition, ranging from ultrabasic to andesitic. All three sites are normally magnetized although only one can be tested for significance. The main mass of West Maui volcano is formed by olivine basalts of the Wailuku Series (1·27 to 1·32 m.y.). Four sites in this Series have significant reversed directions. One site is random and another site has a normal direction but a moderately steep, negative inclination; all other normal sites have shallow, negative inclinations. This latter site is regarded as unreliable and is not considered further. The Wailuku Series is overlain by a veneer of andesites and trachytes forming the Honolua Series (1·15 to 1·17 m.y.). The single site has a random magnetization. A period of erosion preceded the localized eruption of picrite- and nepheline-basalt forming the Lahaina Series, but this was not sampled.

Lanai (Stearns 1940a) consists of a single volcanic dome of olivine-basalt lava. The volcano is exceptional as there are no signs of secondary volcanic activity. All the sites are reversed, but only one can be tested for significance.

Molokai (Stearns & Macdonald 1947) is formed by three volcanoes—Kalaupapa, East and West Molokai. The oldest is West Molokai (1·84 m.y.) and consists of thin bedded olivine-basalt flows. All four sites have reversed, significant directions. East Molokai is the most extensive and has been divided into a lower member (1·47 to 1·49 m.y.) of essentially basaltic composition, and an upper member (1·3 to 1·46 m.y.) of more acidic composition. Four sites were obtained in the upper member, and three in the lower. All sites are significantly grouped except one, and all are reversed, except one. The latter site is ignored as its direction is well away from both normal and reversed groups and samples are thought to be flow banded. The third volcano, Kalaupapa, formed by olivine-basalts, is of small subaerial extent and was not sampled.

Oahu (Stearns & Vaksvik 1935; Stearns 1939, 1940b) is formed of two, or possibly three, volcanoes; Waianae in the west and Koolau in the east. The Waianae volcano basalt flows (2·73 to 3·64 m.y.) have been divided into a lower, middle and upper group, the higher groups tending towards a more acidic composition than the lower olivine-basalts; a Dyke-and-Breccia Complex is also recognized. Radioactive dating (McDougall 1963) of the Mauna Kuwale Trachyte (8·26 to 8·46 m.y.), a rock overlain by a part of the lower member, has led to the suggestion of an older volcano existing on this site. However, this older volcano was not sampled for palaeomagnetic work. The Waianae Series was sampled at 13 sites, most in the Basalt, but some in the Dyke-and-Breccia Complex. Three sites in the Basalt became random after partial demagnetization, while one in the Complex and one

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in the Basalt remain random. Excluding the random sites, the site directions fall into two groups—normal and reversed. Five sites in the Basalt are reversed and two are normal. One site in the Complex is reversed. One site in the Basalt, however, is anomalous in having a reversed declination but a normal inclination, so this site is regarded as unreliable and is discarded. The youngest volcano, Koolau, is divided into three series, Koolau, Kailua and Honolulu, although the Kailua Series is now regarded as the hydrothermally altered equivalent of the olivine-basalts of the Koolau Series (2·14 to 2·56 m.y.). Eight sites were sampled in the Koolau Series and all have reversed directions, although one site is random. The Honolulu Series consists of a group of small, spasmodic eruptions, which followed a long period of erosion, on the south and south-east flank of the Koolau volcano. Two sites were sampled in this series and both have normal significant magnetization.

Kauai (Macdonald *et al.* 1960) is the oldest of the volcanoes sampled and is structurally the most complex. It consists essentially of a single volcanic dome of olivine-basalts forming the Napali formation (4·44 to 5·72 m.y.). Three sites were sampled in this formation and all are normal and significant. The centre of the dome is thought to have collapsed and to have filled with lavas which form the Olokele formation. These lavas eventually overflowed into a graben on the south-west flank, where they constitute the Makaweli formation (3·48 to 4·05 m.y.). A small vent on the south-eastern flank also filled with lavas which constitute the Haupu formation. None of these intermediate formations was sampled. Under-saturated alkali basalts were erupted after a long period of erosion and form the Koloa Series (1·41 to 1·43 m.y.). Two sites were sampled in this Series; one site is normal and significant, but although specimen directions in the other site are not significantly grouped, they are obviously reversed.

3.2 *Average directions and between-site scatter*

In this section, only site directions will be considered which are significant (Section 2) after partial demagnetization in 150 Oersted alternating field. These directions fall into two distinct antiparallel groups, with four exceptions which are regarded as unreliable and are discarded (Section 3.1) as the directions are inconsistent; flowbanding is present at two sites and the other two occur between two opposing polarity zones and probably reflect the direction of the Earth's field during an abnormal period. The normal group ($0°, +35°, \alpha = 10°$) occur in a minimum of three zones, with at least two interjacent reversed zones ($175°, -28°, \alpha = 7°$). These zones appear to be unrelated to broad rock types. The overall mean direction ($357°, +30°, \alpha = 5°$), irrespective of polarity, is similar to the axial dipole field in Hawaii (0°, +37°), although significantly (95 per cent) displaced from it by 2°. The significant displacement of the mean direction from the present geomagnetic field in Hawaii (11·25°, +39·0°) is 10°, and from the inclined geocentric dipole field (12°, +40°) is 11°. Thus although the fit to an axial dipole field is not exact, the fit is much better than to the inclined dipole or present fields in Hawaii.

The estimates of precision of the scatter of site directions about their mean are given (Section 2) by $k = 18$ (circular standard deviation $\psi_{63} = 19·1°$) on the Fisher analysis, and $\beta = 25$ ($\psi_{63} = 17·0°$) on the Watson and Irving analysis. These dispersion estimates may be compared with the geomagnetic (1960 field) scatter around the Earth along the latitude of Hawaii (21° N) and along its paleomagnetic latitude (16° N), which are expressed by $\psi_{63} = 12·7°$ and 14·3° respectively. If the scatters around the same latitudes in both northern and southern hemispheres are averaged

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(i.e., 21° N and 21° S; 16° N and 16° S), then $\psi_{63} = 15{\cdot}5°$ and 17·0° respectively. Similar analyses of the geomagnetic field over the Pacific region, as defined by the Andesite Line, for the same two latitudes give $\psi_{63} = 7{\cdot}8°$ and 8·6°. It is clear that the between-site scatter is much greater than that observed during an axial rotation of the 1960 geomagnetic field in the Pacific, and is slightly greater, but probably not significantly, than that observed during an axial rotation of the complete geomagnetic field.

3.3 *Intensity and susceptibility*

The arithmetic average intensity of NRM of all 303 specimens is $22{\cdot}56 \times 10^{-3}$ gauss, and falls by 25 per cent after partial demagnetization in 150 Oersted alternating magnetic field. The arithmetic average susceptibility (moment induced in 0·5 Oersted divided by the applied field) of 157 specimens, each from a different sample, is $3{\cdot}33 \times 10^{-3}$. However the intensity and susceptibility frequency distributions are skewed and approximate to log-normal distributions. The average intensity, assuming a log-normal distribution, is $5{\cdot}24 \times 10^{-3}$ gauss, with a standard deviation between 1·15 and $23{\cdot}99 \times 10^{-3}$ gauss, and the average susceptibility is $1{\cdot}75 \times 10^{-3}$, with a standard deviation between 0·37 and $4{\cdot}68 \times 10^{-3}$ gauss. Oersted^{-1}.

3.4 *Isothermal remanence*

An isothermal remanence (IRM) was given to 13 specimens from 11 different formations (Roquet 1947). The intensities of maximum IRM were higher than 2 gauss, and could not be measured accurately, and therefore the field required to saturate could not be determined. The coercivity of remanence, however, averaged 445 gauss, ranging between 260 and 700 gauss.

3.5 *Thermoremanent magnetization and anisotropy*

Eight specimens were given a TRM by heating them to 750 °C in a non-magnetic furnace (Irving *et al.* 1961b) and allowing them to cool in the Earth's magnetic field (11·3°, −65·0°, 0·59 Oersted) with their orientation marks, which were randomly oriented with respect to their field positions, in the magnetic meridian. All specimens acquired directions which were parallel, within experimental error, to the applied field. It is evident, therefore, that anisotropic effects, if present, are less than the experimental error in the determination of significant NRM site directions, although some anisotropic effects were suspected in sites which showed flowbanding, but these sites were excluded from the analysis of directions.

The ratio of the TRM intensity acquired in 0·59 Oersted with the initial intensity of NRM was very variable, ranging from 0·07 to 0·46, with an average of 0·28 (standard deviation $\pm 0{\cdot}17$). In view of this variable behaviour, these results cannot be used for determination of the strength of the Earth's magnetic field in the past (Thellier 1938).

3.6 *Magnetic stability*

It is not usually possible to determine the absolute magnetic stability of rock samples (i.e. to determine if the observed direction of NRM is the same as that acquired in the past Earth's magnetic field when the rock originated) but various field and laboratory tests have been devised which can indicate stability. In the case of the Hawaiian data, stability is indicated by four independent criteria.

(1) Partial demagnetization in alternating magnetic fields up to 750 Oersted (Section 2) was carried out on 12 "pilot" specimens chosen to be representative of

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the collection as a whole. The specimens, with three exceptions, behaved very similarly, the initial direction changing in fields less than 50 Oersted to become constant at all higher fields. The pilot specimens behaved very similarly, the directions, with three exceptions, changing slightly in fields less than 50 Oersted to become constant in all higher fields, and the intensity, with two exceptions, showed a slight increase (curves A and B, Figure 3) before a gradual decrease as the applied field was increased. The three exceptions in the behavior of directions, two specimens from the Kula formation and one from the Koolau Series, showed no tendency to group in any particular direction over any range of applied field, and the same two Kula specimens were abnormal in their intensity behaviour (curve C, Figure 3) showing a rapid decrease in low applied fields. The overall behaviour is consistent with the view that the initial NRM is composed of two components, a soft component acquired in the Earth's magnetic field during the last few thousand years, and a hard component acquired when the rock cooled from above its Curie point. The behaviour of the Kula specimens is consistent with the NRM, which is of high intensity, being associated with lightning effects. The NRM of the Koolau specimen is regarded as unstable. Treatment of all specimens in 150 Oersted (peak) alternating magnetic field is thought to remove all secondary low coercivity components, and the exclusion of randomly directed sites (v.i.) should eliminate sites which are either unstable or affected by lightning.

(2) The data, after partial demagnetization, are internally consistent as the directions fall into two antiparallel groups which are divergent from the present Earth's field direction in Hawaii. However, sampling was undertaken so that samples from the same site represent the same short period of time. It is to be expected, therefore, that stable sample directions within the same site will be similar to each

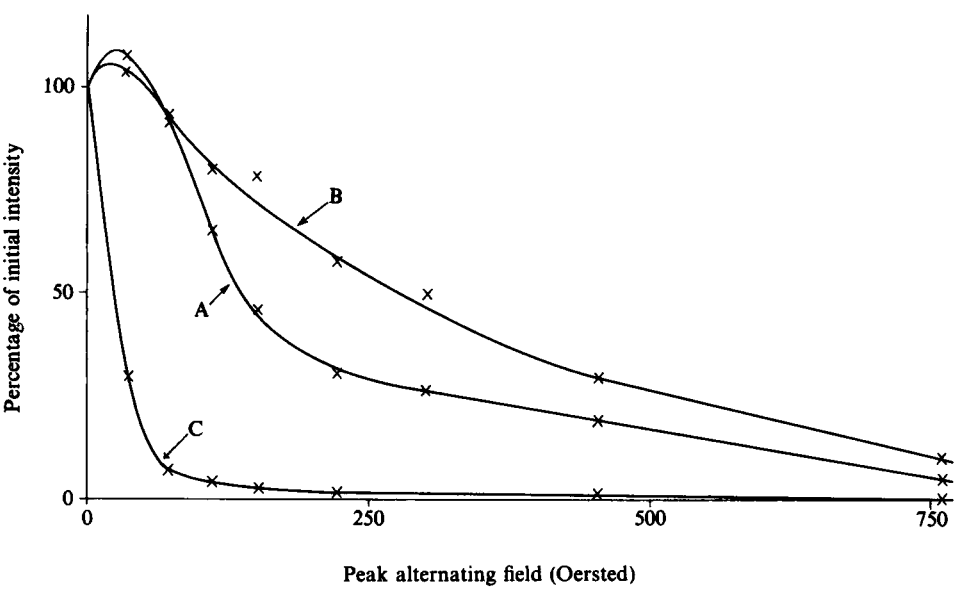


FIG. 3.—Partial demagnetization of NRM–Intensity. Curve A, specimen 8a from the Wailuku formation; and curve B, specimen 89a from the Koloa formation, which are typical of most pilot specimens. Curve C, specimen 42a from the Kula formation which is typical of specimens affected by lightning.

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other. However, twelve of the 59 sites do not have significant mean directions (Section 2), so these sites have been omitted on the grounds of inconsistency, which is probably caused by either instability or lightning effects.

(3) The average coercivity of maximum IRM (Section 3.4), 445 gauss, suggests stability as rocks with high coercivities are generally stable.

(4) There is no obvious evidence of weathering and care was taken to obtain fresh samples. Some of the olivines are partially altered to iddingsite, but this change probably took place during, or immediately after, eruption. However, detailed petrological work is required to confirm that weathering is totally absent.

4. Conclusions

The palaeomagnetic directions are considered reliable indicators of the past geomagnetic field in Hawaii after the samples have been treated in 150 Oersted (peak) alternating field and when sites which have randomly oriented sample magnetization have been excluded. The data can then be considered in terms of the polarity and mean direction of the Earth's magnetic field during the last 5 m.y., and of the scatter of site directions about this mean, attributable to secular changes, which can be used to estimate the maximum magnitude of secular variation during this period.

(a) The presence of normal and reversed polarity may be due to mineral self-reversal (Néel 1955; Verhoogen 1956) or to irregular periodic reversals of the Earth's magnetic field. It is clear, however, that the polarities lie in stratigraphic zones (Table 1) in which the petrology of any one polarity zone does not differ markedly from any other. Furthermore, eight specimens given a TRM in the laboratory (Section 3.5) showed no tendency to self-reversal. Detailed petrological work is still required to test for any significant mineralographic difference between polarity zones (Ade-Hall 1964; Wilson 1964), but it seems likely that the reversals are due to changes in polarity of the Earth's field rather than to self-reversal. Comparisons are necessary with rocks of identical age in different parts of the world before this hypothesis can be confirmed (Cox *et al.* 1963a, b, 1964; McDougall & Tarling 1963, 1964) but it seems reasonable at this stage, to use magnetic polarity as a stratigraphical tool in the Hawaiian islands (Tarling 1962).

(b) The mean direction of the Hawaiian data is similar to other palaeomagnetic data of similar age (Hospers 1955) in showing a mean direction close to that of an axial dipole, although the small significant displacement of the Hawaiian mean direction from the axial dipole direction may be important. However the displacement does not increase with time and does not suggest that there has been large-scale rotation of the Hawaiian islands since their formation, although an east–west translation, as envisaged by Wilson (1963), cannot be tested by this data.

(c) The scatter of site directions about their mean appears to be consistent with the hypothesis that the magnitude of secular variation in the Hawaiian area has not been abnormal during the last 5 m.y. However, the method only affords a maximum estimate and no evidence is available on the periodicity or detailed behaviour of secular variation, for which more detailed sampling is necessary. Doell & Cox (1961, 1963) found that the palaeomagnetic directions between adjacent flows on the island of Hawaii showed very small changes, with a tendency to form a constant direction over a number of flows. This behaviour suggests that short period changes (10^{2-3} years) have not been seen in Hawaii during the last 800000 years, but the data presented here suggests an ordinary magnitude of secular variation over the last 5 m.y. The apparent inconsistency of these two sets of observations

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may be explained by the screening of short period (10^{2-3} years) but the retention of longer period components of secular variation, or by the different age covered by the two sample collections.

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References

Ade-Hall, J. M., 1964. *Geophys. J.*, **8,** 403.
Alldredge, L. R. & Hurwitz, L., 1964. *J. Geophys. Res.*, **67,** 2631.
Cox, A., 1962. *J. Geomagn. Geoelect., Kyoto,* **13,** 101.
Cox, A., Doell, R. R. & Dalrymple, G. B., 1963a. *Nature,* **198,** 1049.
Cox, A., Doell, R. R. & Dalrymple, G. B., 1963b. *Science,* **142,** 382.
Cox, A., Doell, R. R. & Dalrymple, G. B., 1964. *Science,* **143,** 351.
Doell, R. R. & Cox, A., 1961. *Nature,* **192,** 645.
Doell, R. R. & Cox, A., 1963, *J. Geophys. Res.*, **68,** 1997.
Fisher, R. A., 1953. *Proc. Roy. Soc.*, A, **217,** 295.
Fisk, H. W., 1931. *C.R., U.G.G.I., Bull.*, **8,** 280.
Hospers, J., 1955. *J. Geol.*, **63,** 59.
Irving, E., Stott, P. M. & Ward, M. A., 1961a. *Phil. Mag.*, **6,** 225.
Irving, E., Robertson, W. A., Stott, P. M., Tarling, D. H. & Ward, M. A., 1961b. *J. Geophys. Res.*, **66,** 1927.
Lowes, F. J. & Runcorn, S. K., 1951. *Phil. Trans. Roy. Soc.*, A, **243,** 525.
Macdonald, G. A., Davis, D. A. & Cox, D. C., 1960. *Hawaii Div. Hydrography, Bull.*, **13,** 212 pp.
McDougall, I., 1963. *Nature,* **197,** 344.
McDougall, I., 1964. *Bull. Geol. Soc. Amer.*, **75,** 107.
McDougall, I. & Tarling, D. H., 1963. *Nature,* **200,** 54.
McDougall, I. & Tarling, D. H., 1964. *Nature,* **202,** 171.
Néel, L., 1955. *Adv. Phys.*, **4,** 191.
Roquet, J., 1947. *C.R. Acad. Sci., Paris,* **242,**1418.
Runcorn, S. K., 1956. *Handbuch der Physik,* **47,** 470.
Stearns, H. T., 1939. *Hawaii Div. Hydrography, Bull.*, **2,** 75 pp.
Stearns, H. T., 1940a. *Hawaii Div. Hydrography, Bull.*, **6,** 177 pp.
Stearns, H. T., 1940b. *Hawaii Div. Hydrography, Bull.*, **5,** 164 pp.
Stearns, H. T., 1946. *Hawaii Div. Hydrography, Bull.*, **8,** 106 pp.
Stearns, H. T. & Vaksvik, K. N., 1935. *Hawaii Div. Hydrography, Bull.*, **1,** 479 pp.
Stearns, H. T. & Macdonald, G. A., 1942. *Hawaii Div. Hydrography, Bull.*, **7,** 344 pp.
Stearns, H. T. & Macdonald, G. A., 1946. *Hawaii Div. Hydrography, Bull.*, **9,** 363 pp.
Stearns, H. T. & Macdonald, G. A., 1947. *Hawaii Div. Hydrography, Bull.*, **11,** 113 pp.
Tarling, D. H., 1962. *Nature,* **196,** 882.
Thellier, E., 1938. *Ann. Inst. Phys. Globe,* **16,** 157.

1365246b, 1965, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1965.tb03054.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Thellier, E. & Rimbert, F., 1954. *C.R. Acad. Sci., Paris*, **239,** 1399.
Verhoogen, J., 1956. *J. Geophys. Res.*, **61,** 201.
Vincenz, S. A. & Bruckshaw, J. McG., 1960. *Proc. Camb. Phil. Soc.*, **56,** 21.
Watson, G. S., 1956a. *M.N., Geophys. Suppl.*, **7,** 160.
Watson, G. S., 1956b. *M.N., Geophys. Suppl.*, **7,** 153.
Watson, G. S. & Irving, E., 1957. *M.N., Geophys. Suppl.*, **7,** 289.
Wilson, J. T., 1963. *Sci. Amer.*, **208,** 86.
Wilson, R. L., 1964. *Geophys. J.*, **8,** 424.

1365246x, 1965, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1965.tb03054.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License